Structured Approach

- By following a structured approach, you can maximize the chances of success of a data science and a machine learning project at a low cost.
- The structured approach helps the project manager assign the roles and responsibilities of the project team effectively.

Six Steps

- A typical machine learning and data analytics project consists of six steps. The main steps and actions that needs to be taken are as follows.
  - Set the research goal-: The first step is to set a research goal. Every stakeholder understands the what, how, and why of the project. This step also results into a project charter.
  - Data retrieval-: The second step is data retrieval that includes finding the suitable data and getting access to the data from various sources. At the end of the step, we may receive the raw data that needs polishing and transformation.

Six Steps

- Data preparation-: After receiving the raw data, the next step would be to prepare it. The data preparation includes data transformation so that the raw data can be appropriately used in your model. In this step, you detect and correct different kinds of errors in the data, combine data from different sources, and transform it. After the successful completion of this step, you can proceed to the data visualization and data modeling steps.
- Data exploration: - This step helps gain a deeper understanding of the data. You discover the patterns, correlations, and deviations using the visual and descriptive techniques. The insights from this step helps you start modeling.
- Data Model building: - In this step, you gain the insights or make the predictions as stated in the project charter.
- Presenting results and automating the analysis: - This is the last step of the data science and machine learning process. This step helps implement the process changes and automation for better business decisions.

Six Steps

- These steps help achieve the project goals and improve the project success ratio while reducing the project cost.
- This process ensures that you have a well-defined research plan, a good understanding of the business question, and clear deliverables before you start the project.
- In the first step, you focus on getting the high-quality data.
- If you are able to get the high-quality data, the chances of building a high performing data model is high.
- If the data is not of high quality, the chances of building a high performing data model is low because garbage in equals garbage out.

Structured Approach - Benefits

- The structured approach helps you focus on prototype mode where you try multiple models and focus on business value instead of focusing on program speed or coding standards.
- The structured approach also helps the project manager assign the project team on different aspects of the project while focusing on the overall goals.

Structured Approach - Benefits

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- The structured approach also helps the project manager assign the project team on different aspects of the project while focusing on the overall goals.
- The tasks such as extracting and uploading the data to different databases, designing the data scheme that works not only for your applications but also for other projects inside your company, tracking the statistical and data-mining techniques, and expertizing the presentation tools need specialized skills.

Agile Methodology

- Every project does not follow these steps.
- The process followed by a project team is subjected to the preference of the lead data scientist, the company, and the nature of the project.
- Other project models such as **agile** project model is an alternative to the sequential process with iterations.
- Agile project model has gained more ground in the IT department and the companies.
- Hence, the data science communities have also been adopting agile methodologies.

Step 1: Define research goals and create project charter

Define Research Goals

- The project starts with understanding of what, why, and how of the project.
- What is the goal of the project?
- What are the expectations from the project team?
- Why the senior management is investing in the project?
- How the senior management would get the return on investment (ROI) of the project?
- Whether the project is a part of a bigger strategic picture or it is "lone wolf" project.

Project Charter

- The goal of the first phase is answer three questions (what, why, and how) and help the project team and the stakeholders understand what needs to be done and everybody agrees on an action plan.
- The outcome of the step should be a clear research goal, a good understanding of the context, well-defined deliverables, and an action plan with timelines.
- This information is also included in the project charter.

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- This information is also included in the project charter.
- The important tools and techniques adopted during this step are: expert judgement, facilitated workshops, product analysis, alternative generation, questionnaires and surveys, observations, prototypes, benchmarking, and document analysis.
- This stage is guided by a senior project manager or data scientist because this stage requires strong people skills and business acumen.

Goal and Scope of the Project

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- The goal and scope of the project should be clearly documented and signed off by the stakeholders.
- Understanding the project goal and the scope is very important for the success of the project.
- The data scientist need to ask questions continuously until he clearly understands the business expectations and deliverables of the projects.
- He should be clear about how the project fits in the bigger organization objectives, how the project deliverables are going to be used, how the project outcome will change the business.

Goal and Scope of the Project

- After spending months and critical resources, if you realize that you misunderstood the question, you will be frustrated.
- So, this is an important phase.
- Any project manager or data scientist or machine learning engineer should be taken this stage lightly.
- Even though the team is highly motivated and technically competitive, many projects fail to meet their objective, if this stage is not completed appropriately.

Create a Project Charter

- The project charter brings all the stakeholders on the same page and to avoid scope creep, the data scientist should document the problem statement and deliverables, and get the approval from the clients.
- All the information is documented in the project charter.
- The project charter is mandatory for a mid-size to large project.
- The project charter should document the following:
  - The research goals
  - The project mission
  - The project requirements
  - The project scope and deliverables
  - The work breakdown structure (WBS)
  - Resource requirement in terms of man-hours and other item of cost
  - Prototype or proof of concepts (if any)
  - Project schedule
- The project client may need this information to estimate the project cost, resource requirement, and the data required for the project.

Step 2 – Data Retrieval

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Data Retrieval

- During the data retrieval step, the required data is retrieved.
- On many occasions, the data scientist needs to design the data collection process himself.
- On several other occasions, the data is already available within the company.
- If required, the company can also buy the required data i not available in the company.
- These days, high quality data is also available free of cost and commercial use.
- The data scientists also look for such data sources.

Data Retrieval

- The data is available in many forms.
- The objective of the project team is to acquire all the required data which may be a difficult task.
- Even if, you are able to acquire the data, you need to pre-process the data before you can use it.

Acquire Internal Data

- First of all, you should access the relevance and quality of data that is available within the company.
- Many companies adopt a robust data management policy.
- The quality and pre-processed data is readily available in such companies.
- The data is stored in official data repositories such as databases, data marts, data warehouses, and data lakes.

Acquire Internal Data

- The databases are used to store the data.
- Data warehouse is used for reading and analyzing the data.
- A data mart is geared towards serving the needs of a specific business unit.
- The preprocessed data is stored in the data warehouses and data marts whereas raw data is stored in the data lakes.
- In addition to data repositories, data also resides in Excel files on the desktop of a domain expert.

Acquire Internal Data

- Sometimes, the required data is not easily available in the company.
- As the company grows, the data scatters around many places.
- The knowledge of the data also disperses as people change positions and leave the company.
- The metadata is not maintained properly. In such situation, the task of the data scientist becomes challenging.

Acquire Internal Data

- Sometimes, getting access to the data is also challenging.
- Many organizations adopt strict data access policies because they understand the value and sensitivity of the data.
- Due to this, they implement digital and physical barriers given on need-to-know basis.
- Hence, in many organizations, getting access to the data may take time and involve company politics.

Acquire External Data

- If the required data is not available within the organization, the data scientist can look for external data sources.
- Many companies such as Nielsen and GFK specialize in collecting valuable data from various industries.
- Even though the data is very valuable, more and more organizations and governments share their data for free for commercial use.
- Often, this data is of excellent quality.
- The valuable information such as number of accidents or amount of drug abuse in a certain region and its demographics.
- Some of the reputed open-data providers are given below:

Acquire External Data

Open Data Site	Description
Data.gov	US Government's Open Data
https://open-data.europa.eu	European Commission's Open Data
Freebase.org	Open Database that retrieves the information from sites like Wikipedia, MusicBrains, and the SEC archive
Data.worldbank.org	Open Database from the World Bank
Aiddata.org	Open Database for the international development
Open.fda.gov	From the US Food and Drug Administrator

Check the Data Quality at Source

- During the lifecycle of a machine learning or a data science project, the project team spends a good amount of time (up to 80% of the time) in data correction and cleansing.
- The first time when you inspect the data is during the data retrieval stage.
- Most of the errors in data are easy to spot during this stage.
- If the errors are ignored at this stage, you will end up spending good amount of time in solving data issues at a later stage.

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- The data is investigated during three phases of the project lifecycle: data retrieval, data preparation, and data exploratory phases.
- There is a difference in the goal and the depth of investigation during each phase.

Check the Data Quality at Source

- During the data retrieval phase, you check to see if the data retrieved is same as data in the source document and its data types.
- During the data preparation phase, more elaborate data checking is done.
- If you are able to check the data thoroughly during the data preparation phase, you will find errors in the source document.
- You should focus on the content of the variables.
- You should remove typo errors, and convert the data to the common format among the data sets.

Check the Data Quality at Source

- For example, you might correct USQ to USA and Indio to India.
- During the exploratory phase, you focus on the statistical data analysis.
- You assume the data to be clean and look at the statistical properties such as distribution, correlations, and outliers.
- On several occasions, you need to iterate over these processes.
- For example, the root cause of an outlier that you may find in the exploratory phase, could be data entry error. Hence during the process, the data quality improves.

Step 3: Cleansing, Integrating, and Transforming Data

Cleansing, Integrating, and Transforming Data

- During this stage, you sanitize and prepare the data so that it can be used in modeling and reporting phase.
- This is an important step so that performance of the model can be maximized and the strange errors can be minimized.

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- So, the data transformation is also required.
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Sub Processes

- Data Cleansing
  - Data entry errors
  - Redundant whitespace
  - Impossible Values
  - Outliers
  - Missing Values
  - Deviation from a code book
  - Different units of measurement
  - Different levels of aggregation

Sub Processes

- Combine data from various data sources
  - Joining Table
  - Appending Table
  - Using Views
- Data Transformation
  - Reduce the number of variables
  - Creating dummies
  - Extrapolating Data
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Cleansing Data

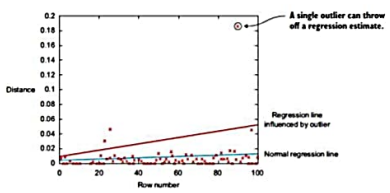
- This process focused on removing the errors from the data so that the data becomes a true and consistent representation of the processes from which it originates.
- At least two types of errors exist in the data.
- The first one is known as interpretation error.
- These are due to values in the data such as saying that a person's age is greater than 300 years.
- Second type of error is due to inconsistencies between the data source and your companies standardized values.

Cleansing Data

- For example, the data source has value "Female" whereas the company standard value is "F" even though they represent the same thing.
- Another example is that the data source has financial values in Dollars whereas the company standard table has financial values in Rupee.
- Sometimes, to identify the data errors, we use advance methods such as simple modeling.
- For example, we used regression to identify the data points that seem out of place.

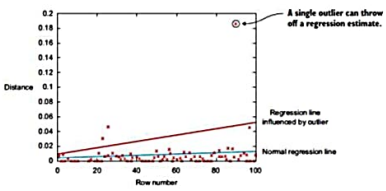
Cleansing Data

- The regression method can also be used to understand the data and detect the influence of each observations on the regression line.
- On various occasion, we may find a single observation that has too much influence.
- Such observations may represent errors or a valid data point.
- Such advanced methods are rarely



Cleansing Data

- The regression method can also be used to understand the data and detect the influence of each observations on the regression line.
- On various occasion, we may find a single observation that has too much influence.
- Such observations may represent errors or a valid data point.
- Such advanced methods are rarely used at the data cleansing stage but they are useful tools.



Data Entry Errors

- The data collection and data entry processes are prone to errors.
- Due to human errors such as typos or lack of concentration, such error occurs.
- The machine generated data is also prone to errors.
- Transmission errors or bugs in the extract are some of the common cause of errors in machine generated data.
- One of the methods of check the manual data entry errors is to tabulate the data with counts.
- For example, if a variable in the dataset can take only two values: "Good" and "Bad".
- You can create a frequency table to check if only those two values are present. In the table below, the values "Goood" and "Bda" point out wrong entries. At least in 20 cases
- Such errors can be fixed manually or programmatically.

Data Entry Errors

Value	Count
Good	1678763
Bad	1274648
Goood	15
Bda	5

Redundant Whitespace

- Whitespaces at the end of the string are hard detect and results in wrong results from the data analysis.
- During the ETL process, the whitespace may lead into errors in the dataset.
- For example there is a difference between "AB " and "AB".
- During the data cleaning process, most of the programming languages can easily remove the leading and trailing whitespace.
- In Python, the leading and trailing whitespaces can be removed by using strip() function.

Capital Letter Mismatch

- The capital letter mismatches in data are common.
- For most programming languages there is difference between "India" and "india".
- Using Python you can solve this problem by applying a function, .lower(), that returns both strings in lowercase.

"India".lower() == "india".lower()

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“India”.lower() == “india”.lower()

- should result in true.

Impossible Values

- Another valuable type of data check is sanity check of impossible values.
- In this case, you need to check whether there is any physically or theoretically impossible values in the data set.
- People taller than 3 meters or people with an age of 299 years are examples are impossible values.
- You can do the sanity check with rules: check = 0 <= age <=120

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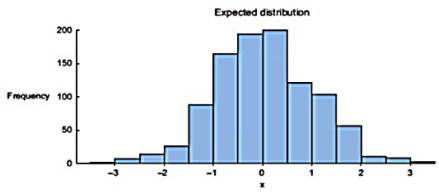
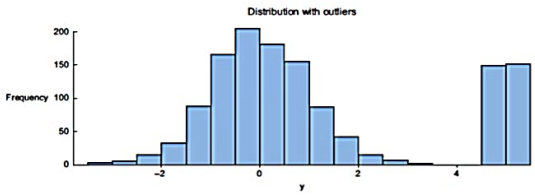
Outliers

- The observation that seems to be distant from other observations is known as outliers.
- Outliers follow different generative process or logic than rest of the observation.
- You can easily find the outliers using a plot or a table with the minimum and maximum values.
- The first figure below shows no outliers whereas the second figure shows the presence of the outliers.





Outliers

- The first plot shows that the most of the observations are around the mean value of the distribution.
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Outliers

- The first plot shows that the most of the observations are around the mean value of the distribution.
- The frequency of observation decreases as they are away from the mean value.
- For a normal distribution, the high values in the right hand side of the bottom graph shows the presence of the outliers.
- Outliers can influence the results of your data model.
- So, you need to investigate them first.

Missing Values

- In our dataset, you may find several missing values.
- They are not always wrong, still you need to handle them.
- Several data modeling techniques cannot handle missing values effectively.
- Sometimes, they may be a result of faulty data collection process or an error during ETL process.
- The common techniques of fixing the missing values are given below.
- The data scientists and machine learning engineers use them according to their data and requirements of the model.

Missing Values

Technique	Advantage	Disadvantage
Omit the values	Easy to perform	You lose informati
Set value to null	Easy to perform	All the data mo handle the null va
Insert a static value such as 0 or the mean	Easy to perform Information from other variables in the observation is not lost	May result into the false estimations from the model
Insert values based on estimated or theoretical distribution	Model is not disturbed	Harder to execute You make data assumptions
Modeling the value (nondependent)	Model is not disturbed	Can lead to too much confidence in the model Can artificially raise dependence among the variables Harder to execute You make data assumptions

Unit of Measurements

- While integrating two different data sets, you should check the respective unit of measurement.
- For example: the distance between two cities in the world can be provided by different data providers.
- One dataset may contain the distance in miles where are other dataset may contain the distance in kilometers.
- You can use the simple conversion formula to fix such issues.

Correct Errors as early as possible

- Many key organization decisions of depend on the insights from the data.
- Hence organizations spend good amount of money on the data retrieval process.
- But the data retrieval process is a difficult, expensive, and error prone task.

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- Many key organization decisions of depend on the insights from the data.
- Hence organizations spend good amount of money on the data retrieval process.
- But the data retrieval process is a difficult, expensive, and error prone task.
- Hence, as soon as the data is retrieved, is should be corrected due to following reasons.

Correct Errors as early as possible

- Spotting the data anomalies is not easy for everyone. The organization may end up in taking costly and wrong decisions, if they are taken based models that are using the incorrect data
- If the data cleansing is not done at an early stage, all the downstream projects have to clean the data that is a costly and time-consuming process
- Data errors may be due to a fault in the business process that needs to be corrected else it may result in the loss of revenue for the organization
- Data errors may be due to defects in the equipment such as defective sensors
- Data errors may be due to bugs in the software that may be critical for the company.

Correct Errors as early as possible

- Ideally, the data errors should be fixed as soon as it is captured.
- But the data scientists do not control every data sources.
- They may point out the data errors but it is up to the owner of the data to fix the issue.
- If you cannot fix the data at the source, you have to handle it inside your data model and the code.
- It is always recommended to keep a copy of the original data.
- On several occasions, when you start data cleaning process, you may make mistakes.
- You may insert the wrong values, or delete an outlier that contained useful information, or alter the data due to misinterpretation of the data.

Correct Errors as early as possible

- If you have a copy of the original data, you can start again.
- The real time data is manipulated at the time of arrival.
- Hence it is not possible to keep a copy of the original data.
- In that case you can use data between certain timeframe for tweaking before you start using the data.
- Cleansing the individual data is not the most difficult thing.
- However, combining the data from different sources is the real challenge.

Combine Data from Different Sources

- In this step we focus on integrating the data that we get from different sources.
- This data varies in size, type, and structure.
- It is available in several formats ranging from database and Excel files to text documents.
- In this section for the sake of brevity, we will focus on data in table structure.
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- In order to combine data from two different sources, you can perform two operations: joining and appending or stacking.
- In the joining operation, we enrich an observation from one table with information from another table.
- In the second operation, appending or stacking, we add the observations of one table to those of another table.
- During these operations, you can either create a new physical table or a new virtual table, called view. The views do not consume much of the disk space.

Joining Tables

- Joining tables help you combine two corresponding observations in two tables and enrich a single observation.
- For example, if one table contains the information about the purchases of a customer.
- Another table contains the information about the region where the customer lives.
- You can join the tables to combine the information and use it for your model.

Joining Tables

- In order to join table, you need to use the columns or variables that represent the same observation in both the tables such as customer name, customer id, or social security number.
- These common fields are known as keys.
- When the keys store unique and not null information, they are called Primary Keys.
- In the picture above, you can see that Client variable is common in two tables.
- Using the join, the region variable has been included with the purchase information.

Client	Item	Month
John Doe	Coca-Cola	January
Jackie Qi	Pepsi-Cola	January

Client	Region
John Doe	NY
Jackie Qi	NC

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